

Design of Hybrid Steel Fibre Reinforced Concrete Beams for Flexure

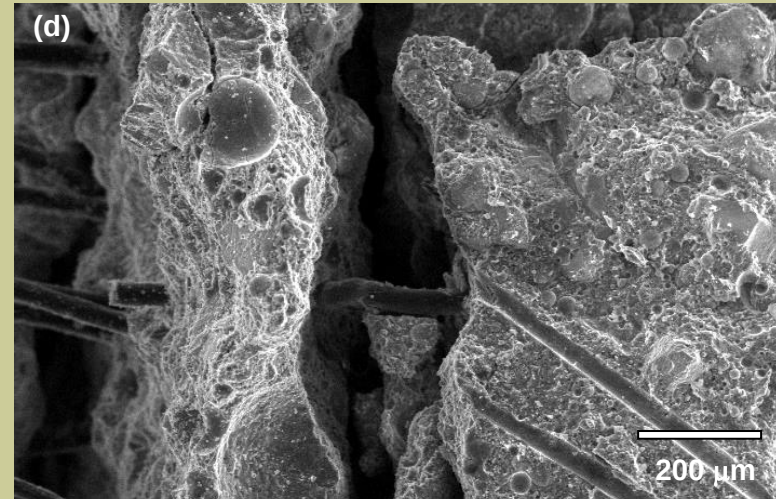
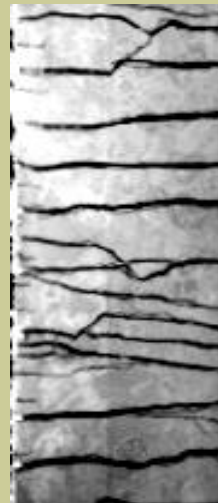
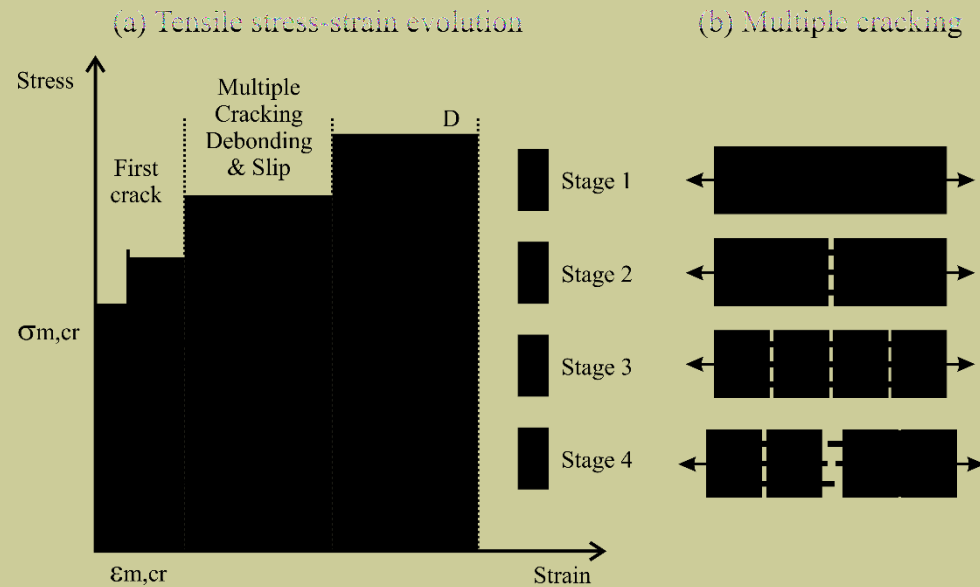
CEM Seminar 2019
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Materials Characteristics

- Ductility
- Toughening
- Improved tensile strength
- Increase level of energy absorption
- Fatigue life, impact/explosive loading conditions, and seismic resistance
- Steelwork, labor, construction time.
- Corrosion damage
- Long-term repair and maintenance.



Modelling Approach

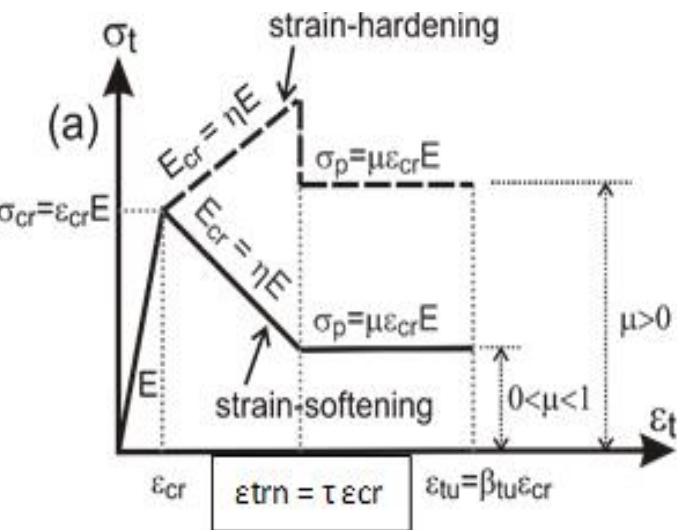
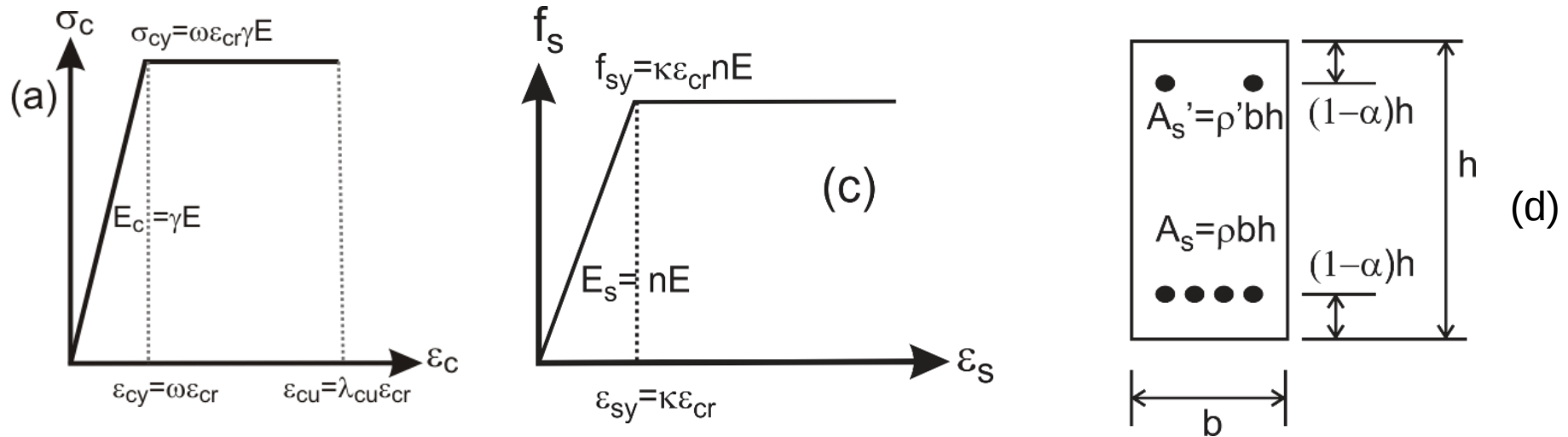
■ Materials evaluation

- Parametric linearized materials model
- Cross sectional analysis based on stress and strain diagrams
- Closed-form solutions of moment and curvature responses
- Back calculation of tensile responses and extraction of tensile properties

■ Structural response check

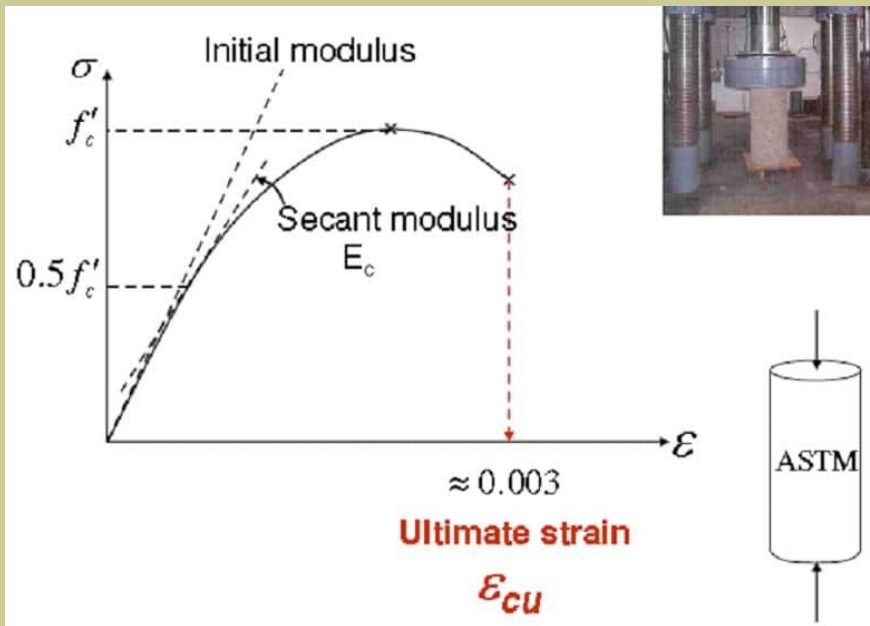
- Forward simulation of flexural load-deflection responses
- Check of parameters at serviceability states: curvature, deflection, post-crack stiffness
- Design recommendations: moment capacity, minimum reinforcement ratio, serviceability limits

Material Models

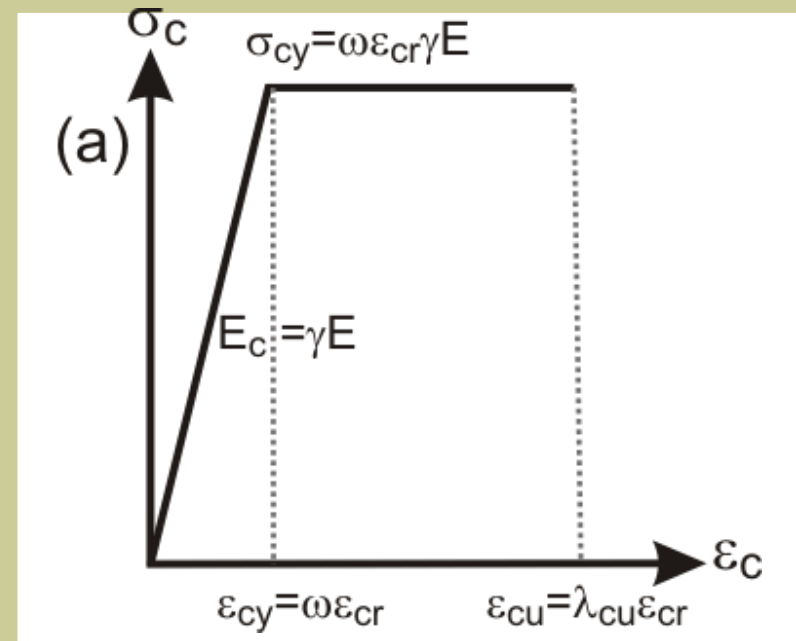


- (a) Elastic-Perfect Plastic Model for Compression,
- (b) Multilinear Tension Model,
- (c) Rebar Model,
- (d) Rectangular Cross Section.

Compressive stress-strain curve

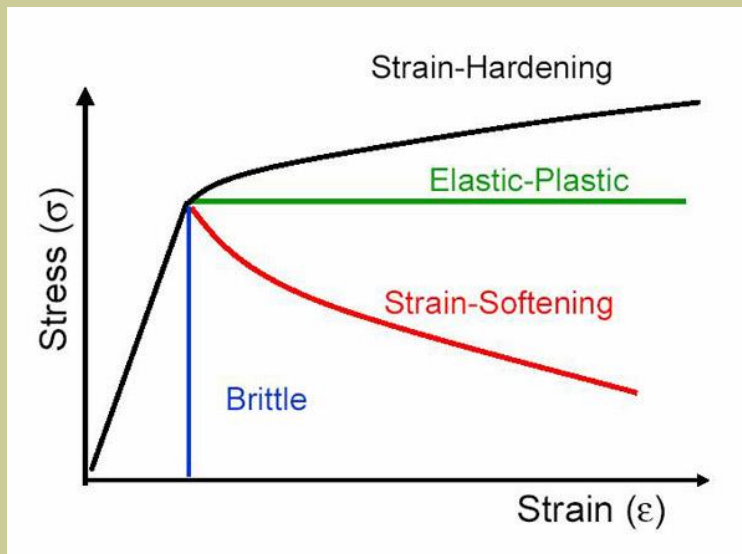


Actual curve

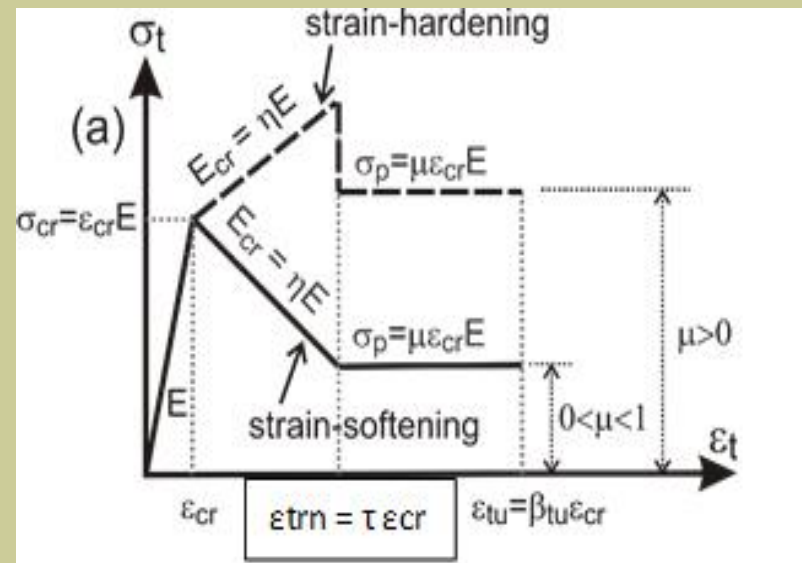


Idealize curve

Tension stress-strain curve

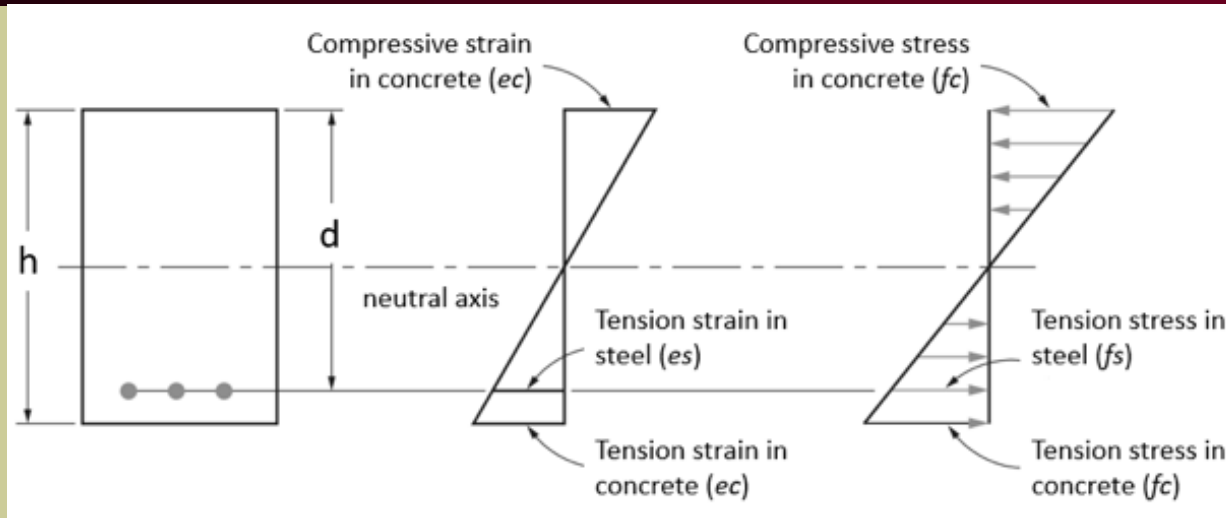


Actual curve



Idealize curve

Basic Flexural behavior and assumption of ACI 318



Assumption to calculate moment capacity

- 1) Strain in steel and concrete are considered linear that being proportional to the distance from neutral axis.
- 2) Concrete zone are considered to reach their flexural capacity when strain equal to 0.003.
- 3) Tensile in concrete below neutral axis is neglected.
- 4) Stress in reinforcement is linear until meet the specified yield strength.
- 5) Compressive stress distribution are performed by a rectangular stress distribution

Basic Flexural behavior and assumption of ACI 318

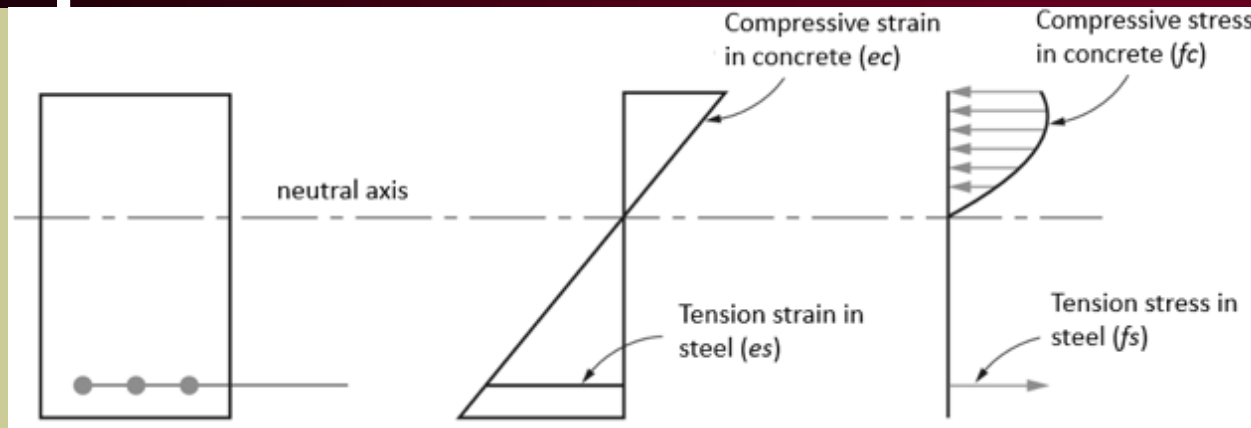


Figure a) Actual stress distribution in concrete

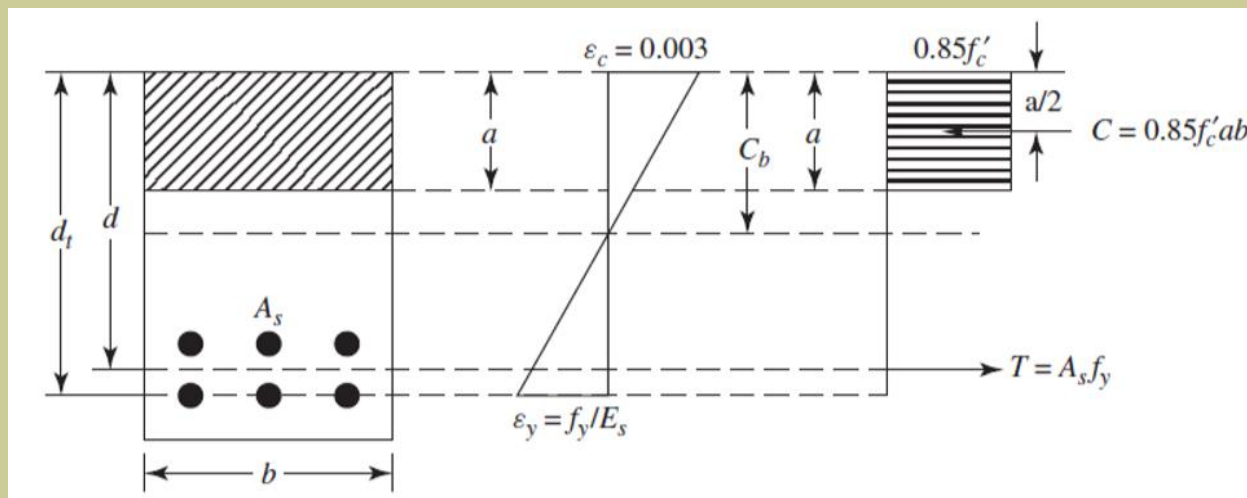
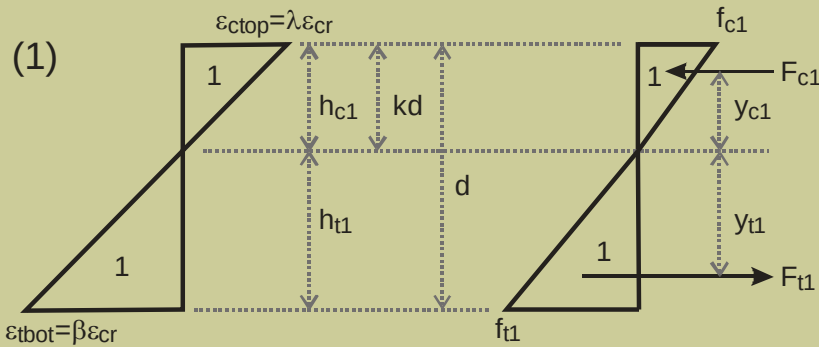


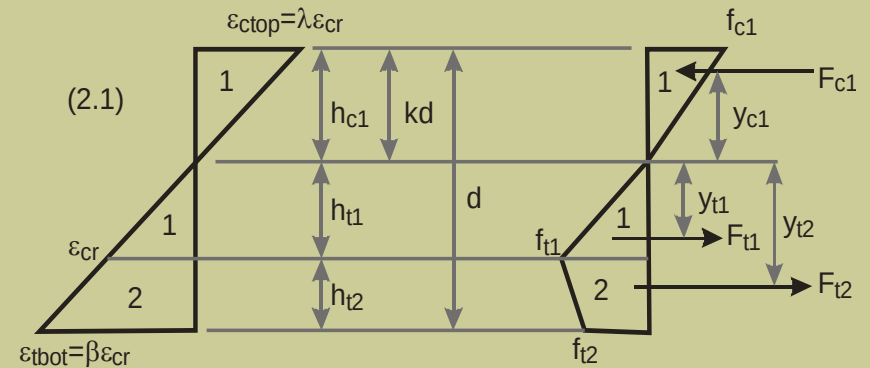
Figure b) ACI318 stress distribution in concrete

Cross Sectional Analysis with ASU closed-form solution for FRC

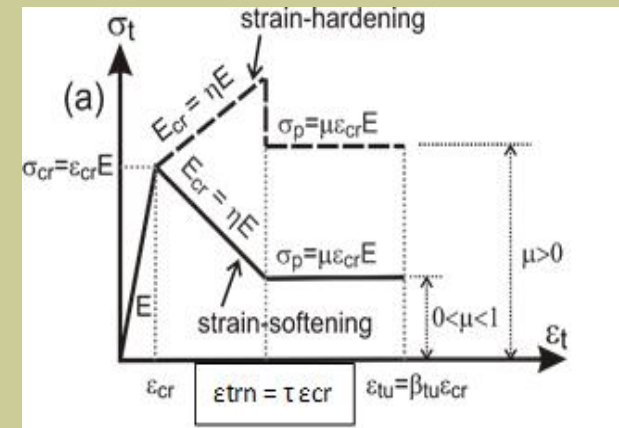
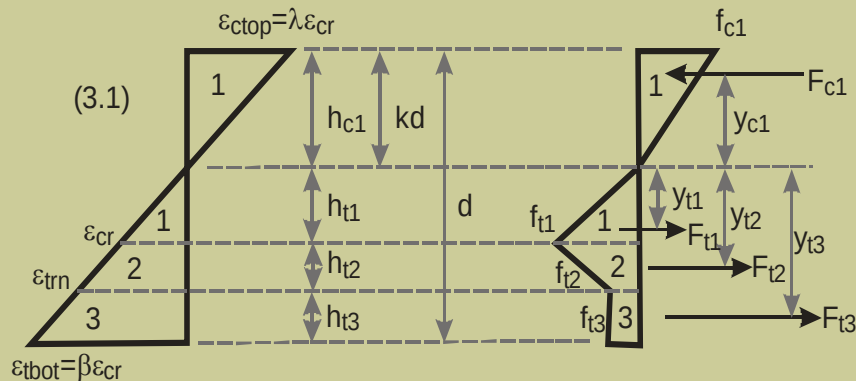
$$0 < \beta \leq 1 \text{ and } \lambda \leq \omega$$



$$1 < \beta \leq \alpha \text{ and } \lambda \leq \omega$$

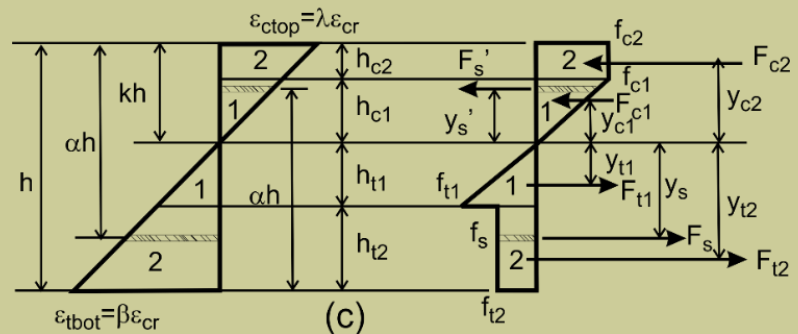
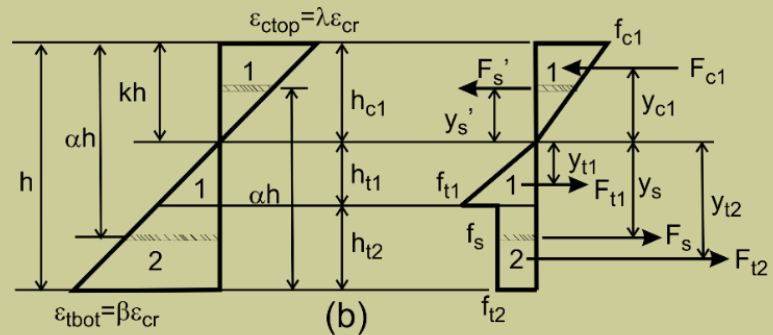
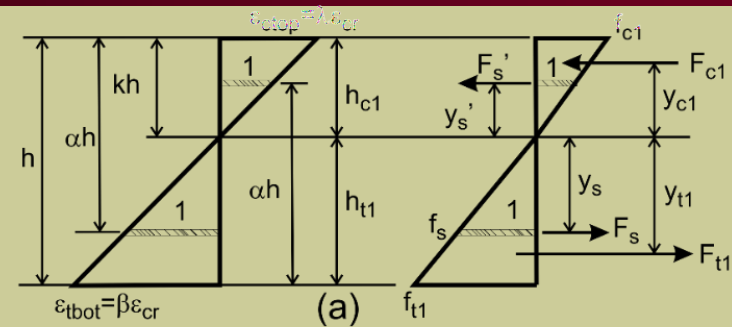


$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda \leq \omega$$



Cross Sectional Analysis with ASU closed-form solution for HRC

Stress diagrams
Hybrid RC
With tri-linear model

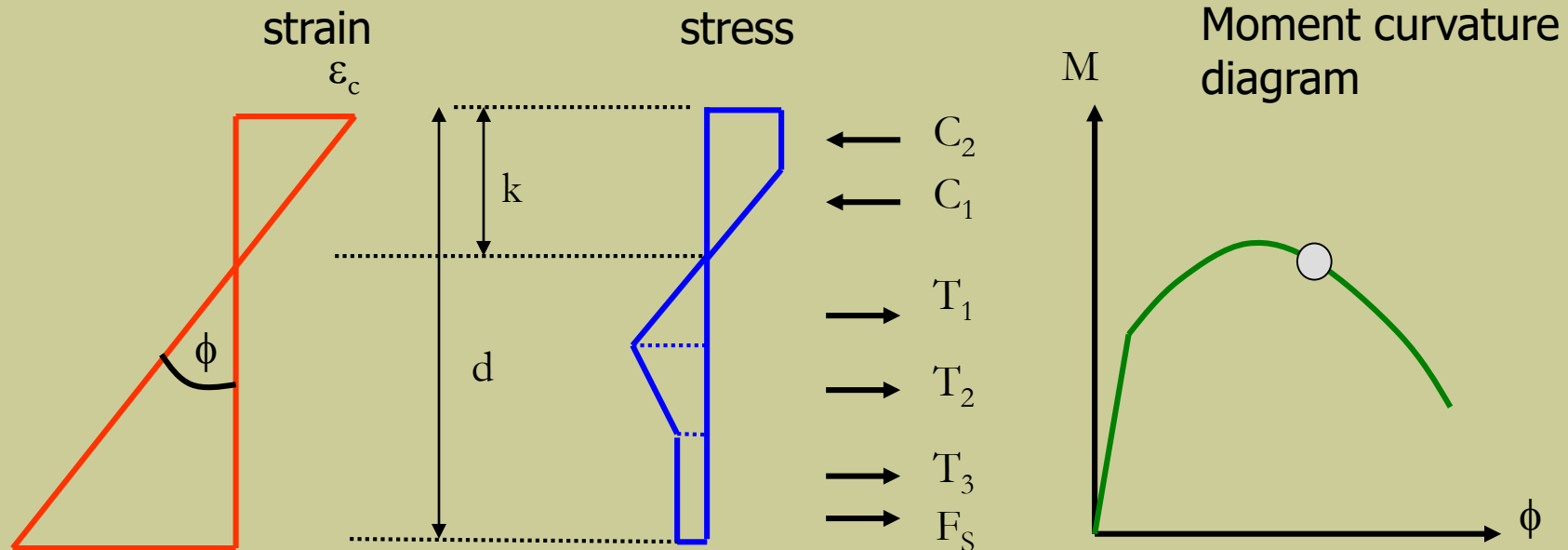


Moment-Curvature Diagram

- Incrementally impose $0 \leq \varepsilon_t \leq \varepsilon_{tu}$
- Strain Distribution
- Stress Distribution
- $\Sigma F = 0$, determine k (Neutral axis)
- $M = \Sigma C_i y_{ci} + \Sigma T_i y_{ti}$ and $\phi = \varepsilon_c / kd$
- Normalization $M' = M / M_0$ and $\phi' = \phi / \phi_{cr}$

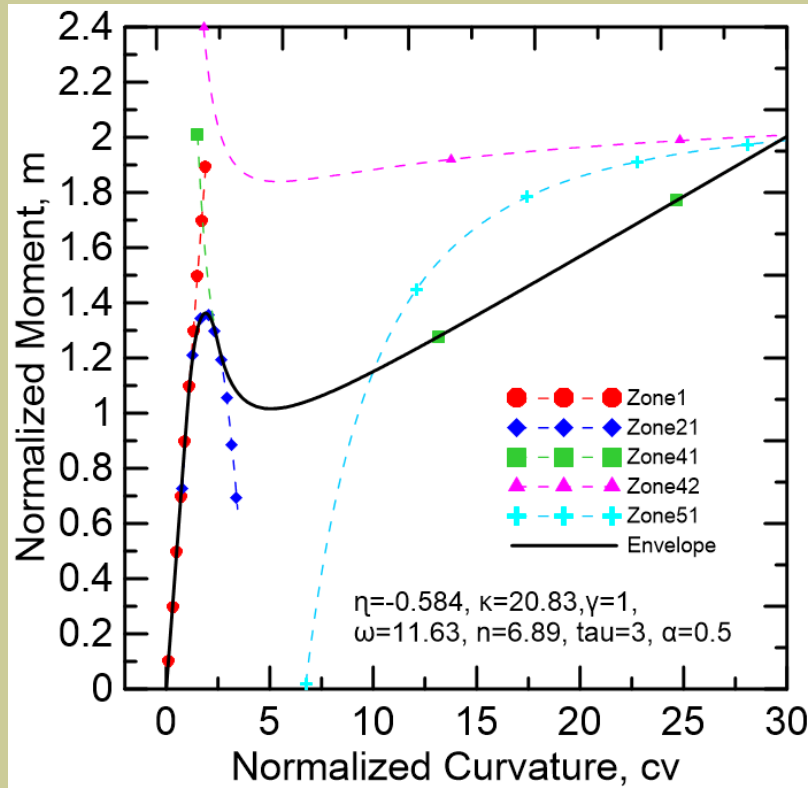
$$F_{c1} = b \int_0^{kd} f_{c1}(y) dy$$

$$y_{c1} = \frac{b}{F_{c1}} \int_0^{kd} f_{c1}(y) y dy$$

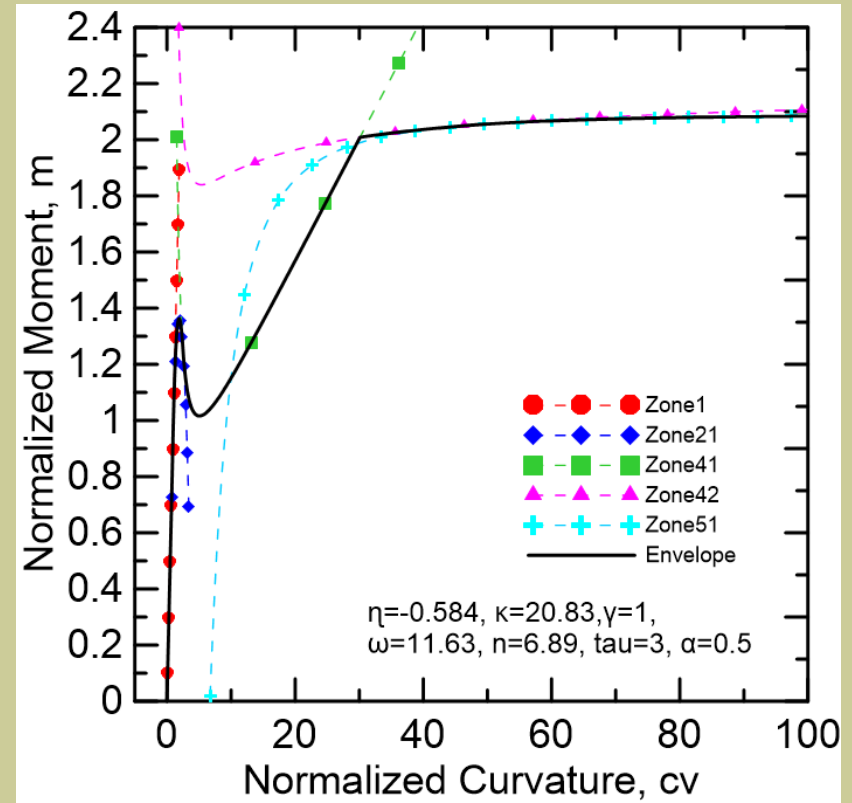


$$0 \leq \varepsilon_t \leq \varepsilon_{tu}$$

Moment-Curvature Diagram



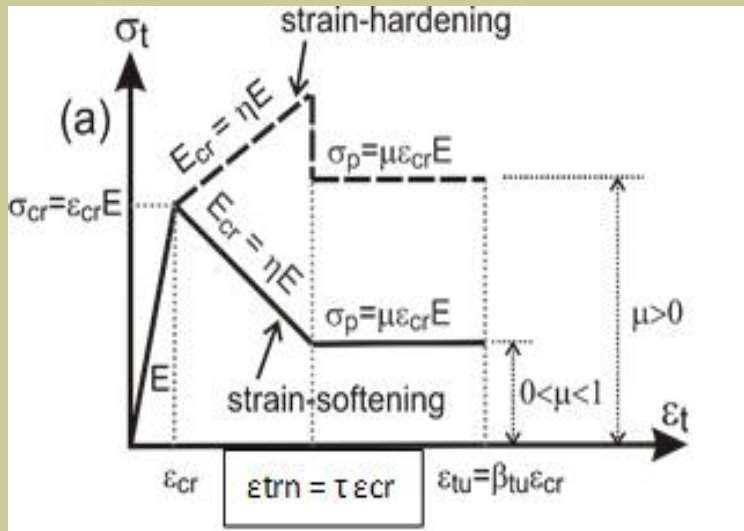
- Example of moment-curvature interactions in the range of 0-30 Normalized Curvature



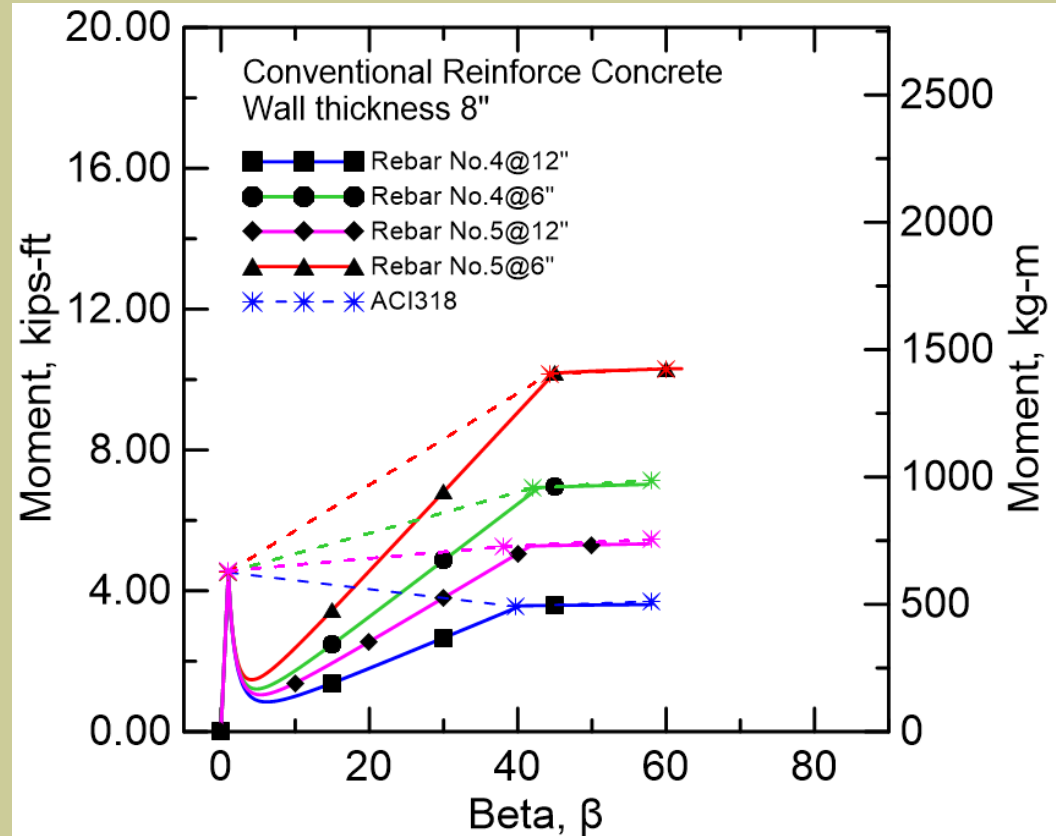
- Example of moment-curvature interactions in the range of 0-100 Normalized Curvature

Verified model with ACI318

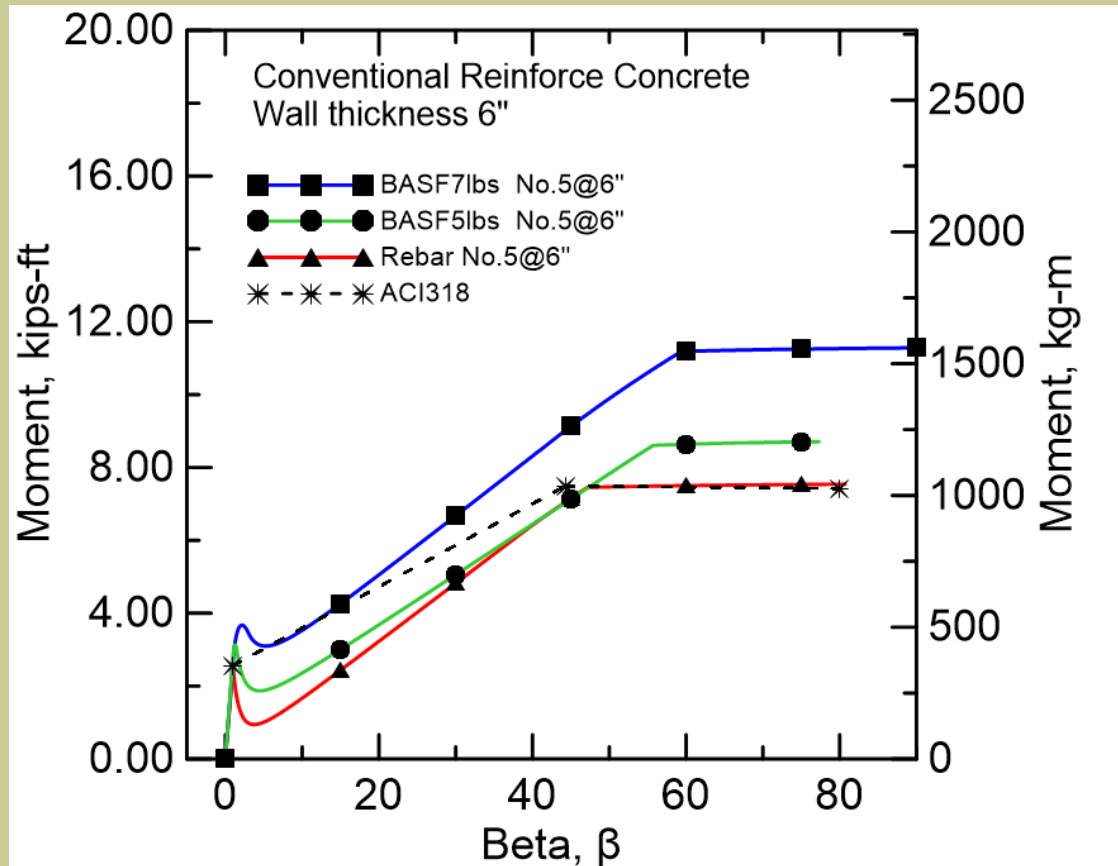
ASU Hybrid Model and ACI318 method



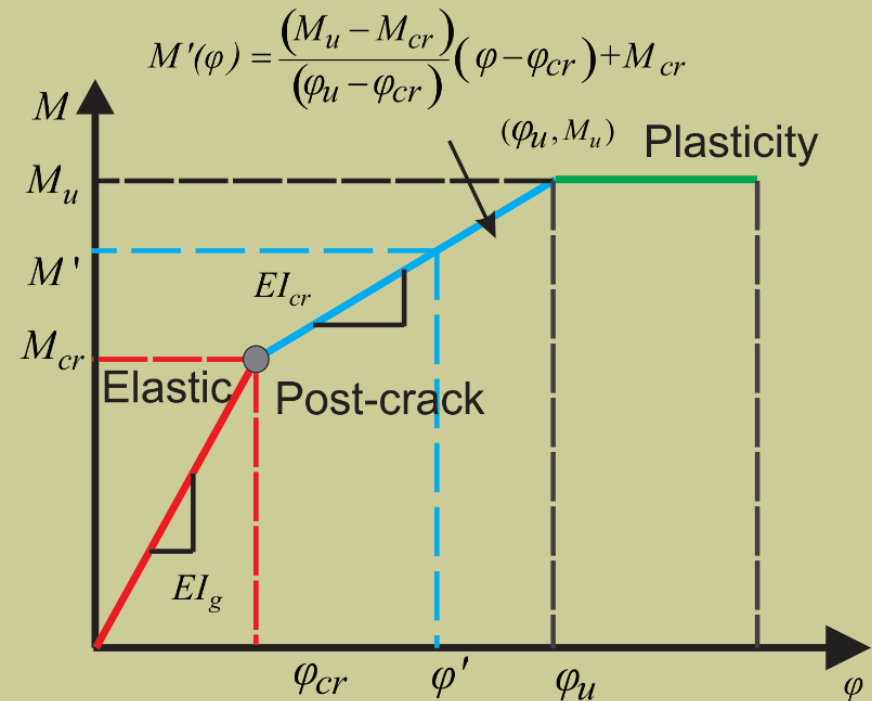
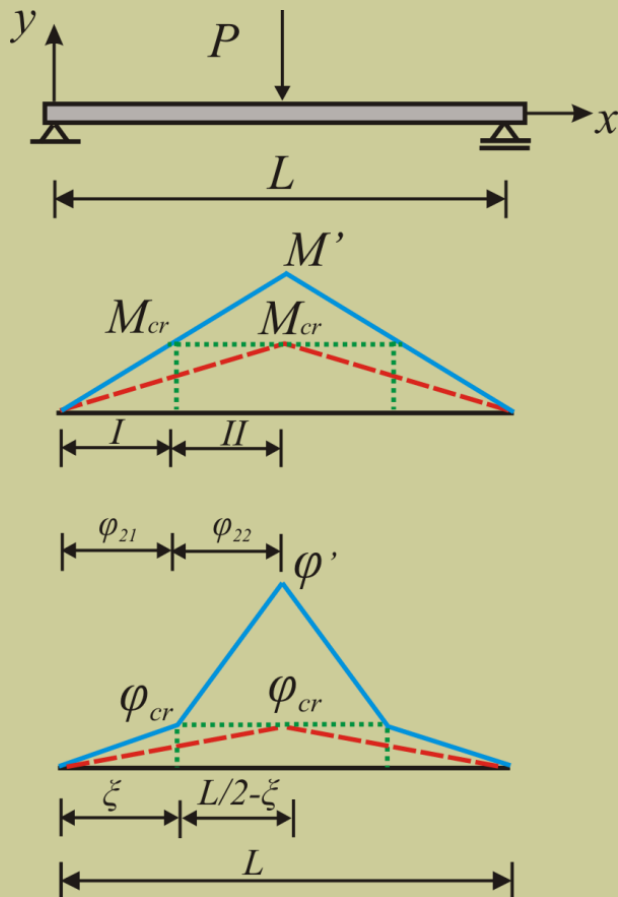
$\tau = 0$, $\mu = 0$, $\eta = \text{undefined}$



RC and HRC comparison

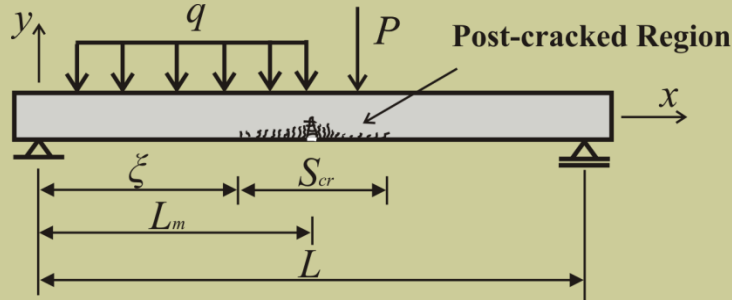


Moment & Curvature Distributions Due to equilibrium



- Use Static Equilibrium to get moment distribution
- Curvature distributions along the beam are generated based on simplified bilinear moment-curvature model

From Moment-Curvature to Load-Deflection



➤ Rotation

$$\text{Region I : } \theta_{21} = \int_0^x \phi_{21}(x) dx + C_1 \quad 0 \leq x \leq \xi$$

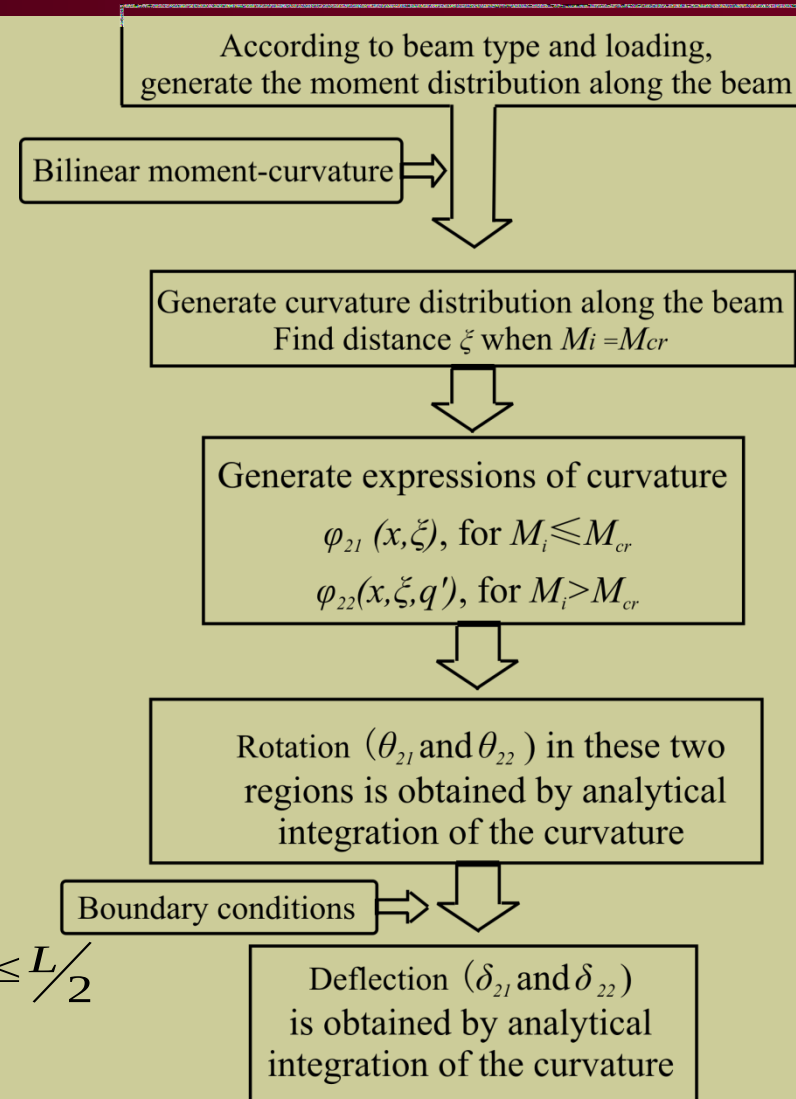
$$\text{Region II: } \theta_{22} = \int_0^{\xi} \phi_{21}(x) dx + \int_{\xi}^x \phi_{21}(x) dx + C_2 \quad \xi \leq x \leq L/2$$

➤ Deflection

$$\text{Region I : } \delta_{21} = \int_0^x \theta_{21}(x) dx + C_3 \quad 0 \leq x \leq \xi$$

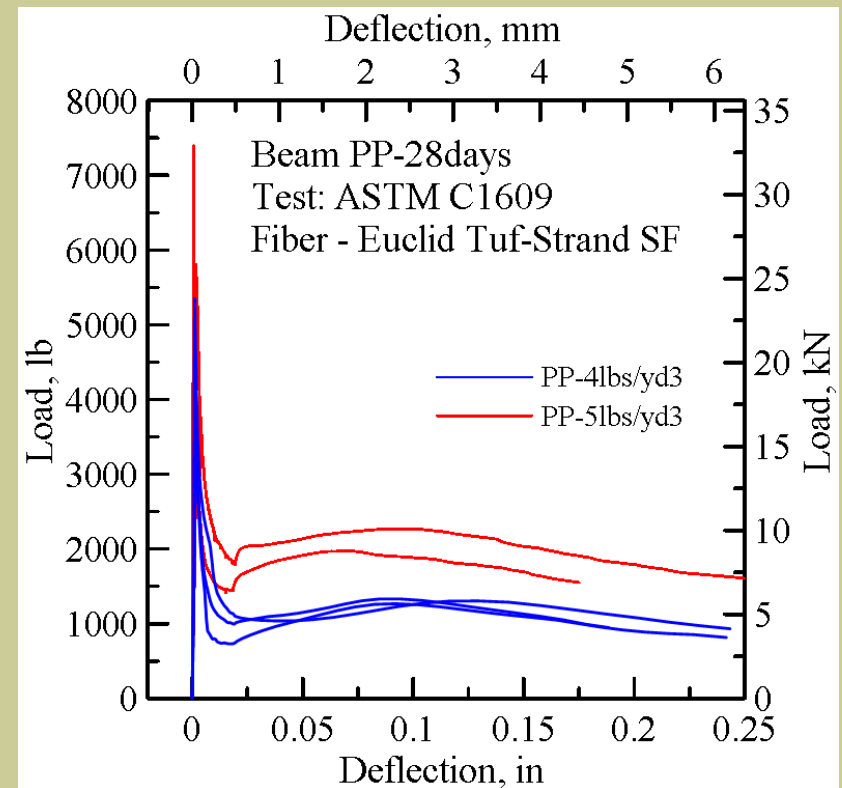
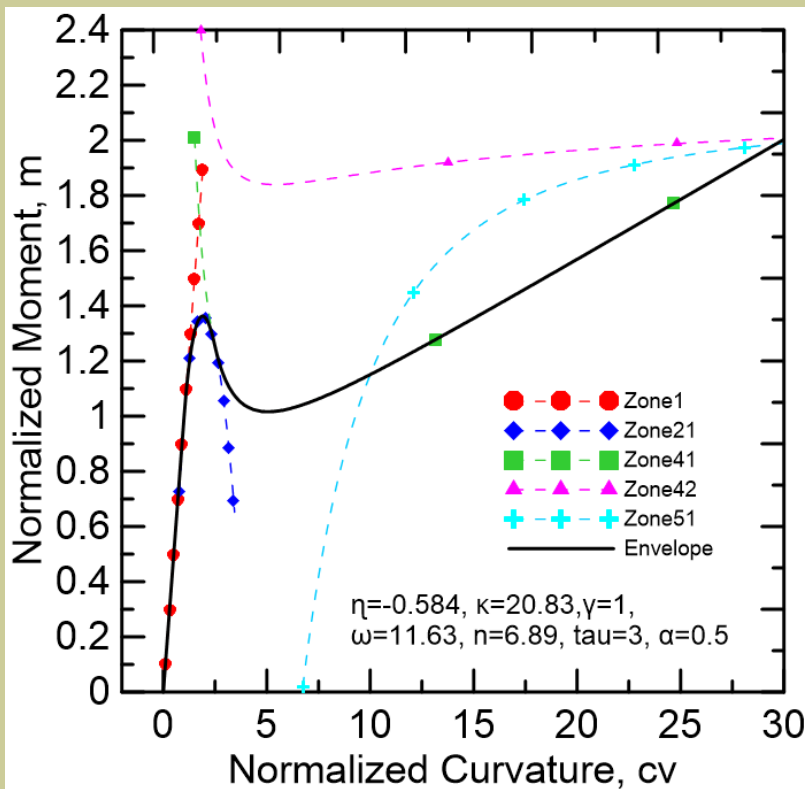
$$\text{Region II } \delta_{22} = \int_0^{\xi} \theta_{21}(x) dx + \int_{\xi}^x \theta_{22}(x) dx + C_4 \quad \xi \leq x \leq L/2$$

The constants of integration defined in terms of C_i are numerical values which are determined based on the boundary conditions.

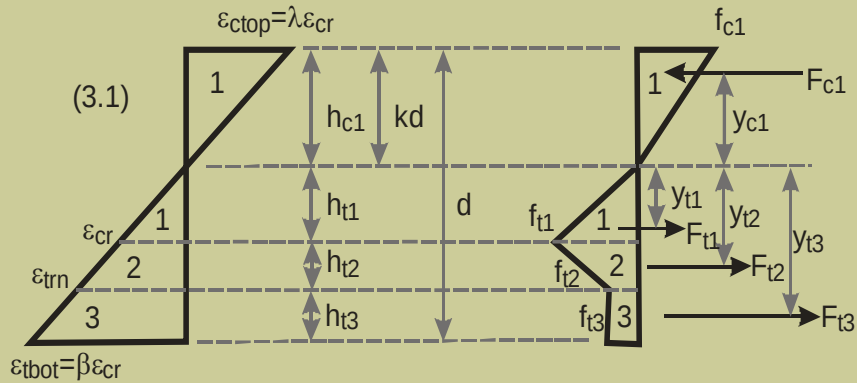


What is the most challenging

- How can we get the accurate parameters from lab testing?

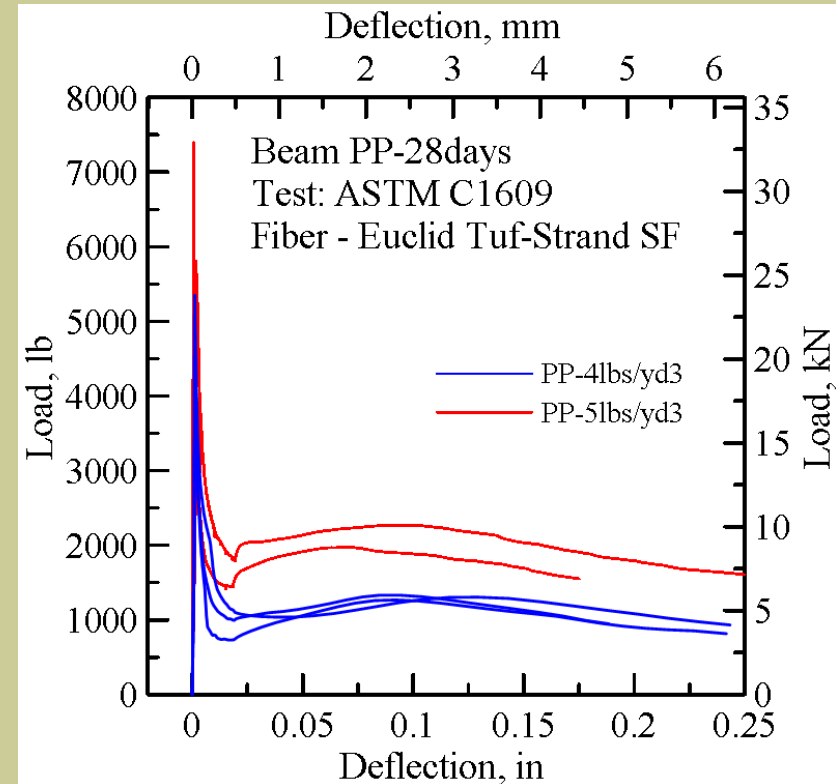
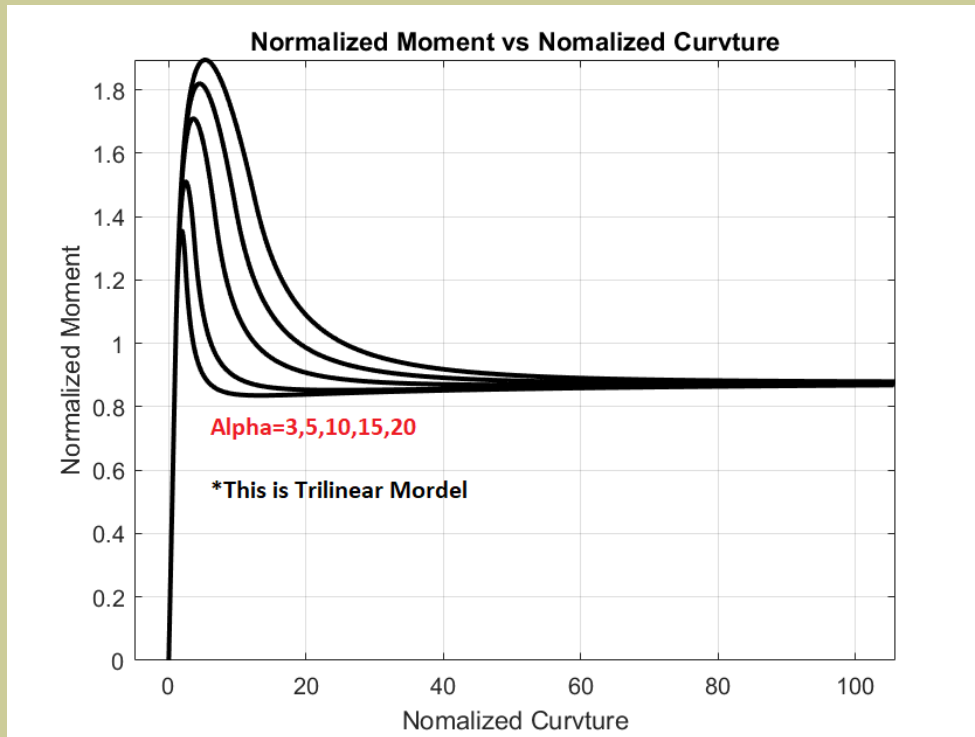


What is the challenging ?

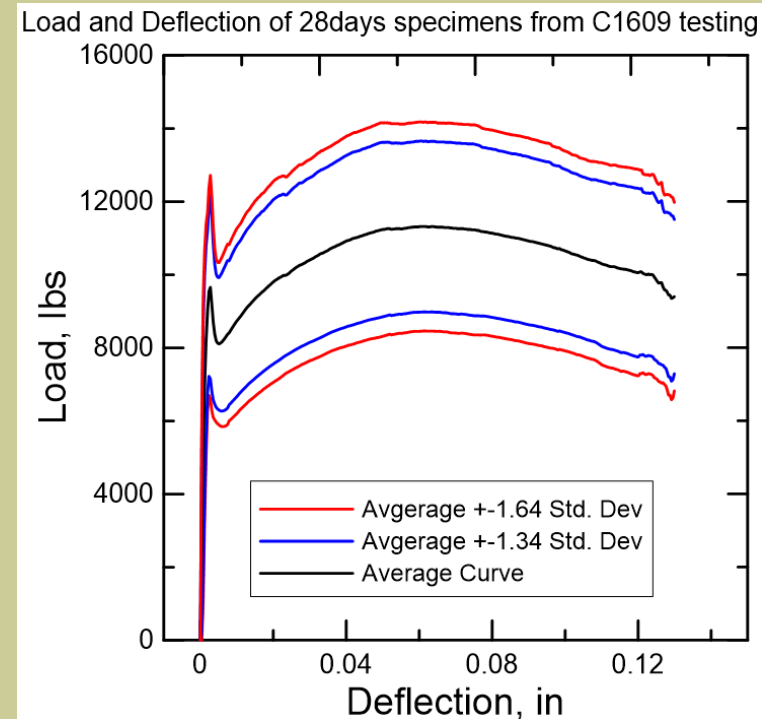
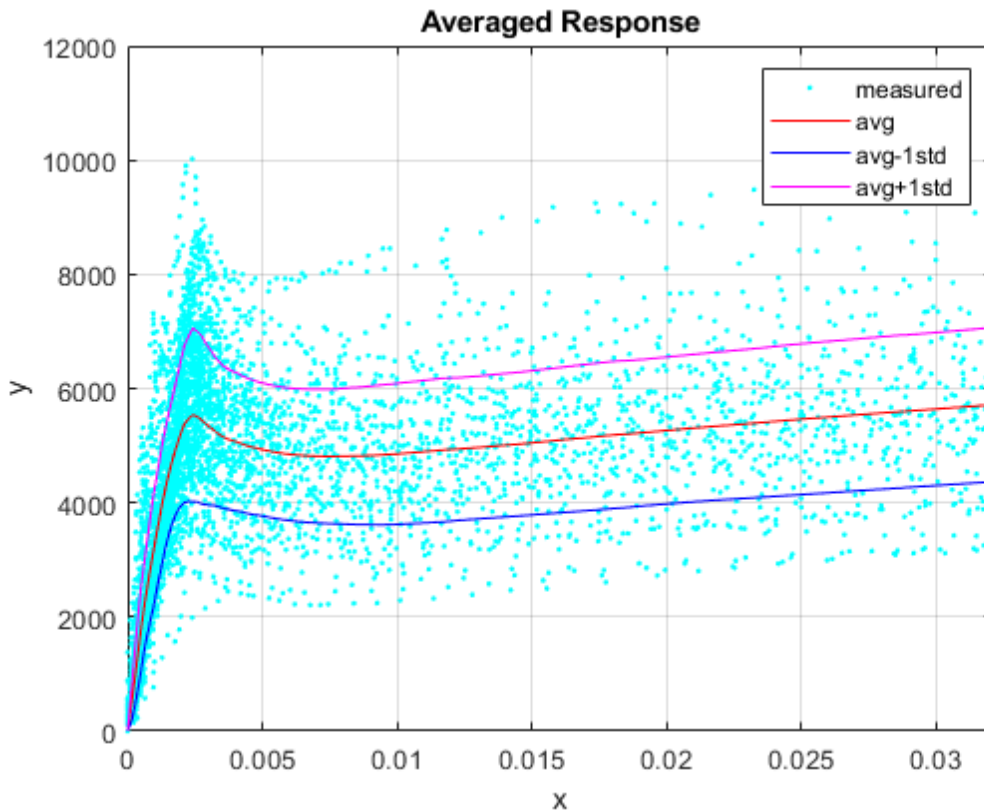


$$\frac{1}{\beta^2 (-1 + k)} \left((3 \xi \omega \beta^2 - \xi \omega^3 - 3 \mu \tau^2 - 3 \eta \tau^2 + \eta + 3 \tau^2 + 2 \eta \tau^3 - 1 + 3 \mu \beta^2) k^3 + (-9 \tau^2 - 3 \eta + 9 \mu \tau^2 - 3 \xi \omega \beta^2 - 6 \zeta \rho n \beta^3 + 3 \xi \omega^3 - 6 \eta \tau^3 + 9 \eta \tau^2 - 9 \mu \beta^2 - 6 \rho n \kappa \beta^2 + 3) k^2 + (-12 \zeta \rho n \beta^3 \alpha + 9 \mu \beta^2 + 6 \rho n \kappa \beta^2 \alpha - 3 + 9 \tau^2 + 6 \rho n \kappa \beta^2 - 3 \xi \omega^3 + 3 \eta + 12 \zeta \rho n \beta^3 - 9 \eta \tau^2 - 9 \mu \tau^2 + 6 \eta \tau^3) k + 1 + 3 \mu \tau^2 + 3 \eta \tau^2 - 6 \zeta \rho n \beta^3 - 6 \rho n \kappa \beta^2 \alpha + \xi \omega^3 + 12 \zeta \rho n \beta^3 \alpha - 6 \zeta \rho n \beta^3 \alpha^2 - \eta - 3 \tau^2 - 3 \mu \beta^2 - 2 \eta \tau^3 \right)$$

One of idea for curve fitting



One of idea for curve fitting



The next step

- Analytical deflection equations of strain hardening and softening fiber reinforcement concrete beam double reinforce concrete will be developed based on simplified trilinear moment-curvature models.



- Composite section will be study with the same model with HRC
- Accuracy of the closed-form deflection equations will be identified by comparing the simulated data with large scale experimental data.

Question ?